

BICAFAR 1G STERILE**DESCRIPTION:**

BICAFAR 1G STERILE: White or slightly yellowish powder; constituted solution is colourless or straw yellow.

COMPOSITION: Each vial contains 1 g of Cefoperazone (as Cefoperazone Sodium).

PHARMACODYNAMICS:

Cefoperazone sodium has similar antimicrobial activity to ceftazidime and cefotaxime. Cefoperazone has a bactericidal action similar to cephamandole, but a broader spectrum of activity. It is highly stable to hydrolysis by most beta-lactamases and has greater activity against Gram-negative bacteria than first- or second-generation cephalosporins. However, it generally has slightly less activity than first-generation cephalosporins against Gram-positive organisms. Cefoperazone has similar antimicrobial activity to ceftazidime although it is slightly less active against some Enterobacteriaceae. It has good activity against *Pseudomonas aeruginosa*, but is less active than ceftazidime.

PHARMACOKINETICS:

Cefoperazone is administered parenterally as the sodium salt. After intramuscular administration of doses equivalent to cefoperazone 1 to 2 g, peak plasma concentration of 65 and 97 µg per mL have been reported after 1 to 2 hours. The half-life of cefoperazone is about 2 hours, and 82 to 93% in the circulation is bound to plasma proteins. Cefoperazone is widely distributed in body tissues and fluids. It diffuses across the placenta, and is excreted in breast milk in low concentrations. Cefoperazone is excreted principally in the bile where it rapidly achieves high concentrations. Up to 30% of a dose is excreted in the urine within 12 to 24 hours. Urinary excretion is primarily by glomerular filtration and this proportion may be increased in patients with hepatic or biliary disease. Cefoperazone penetration into CFS is generally poor.

INDICATIONS:

Cefoperazone is indicated for the treatment for the following infections when caused by susceptible organisms.

Respiratory tract infections (upper and lower); urinary tract infections (upper and lower); peritonitis, cholecystitis, cholangitis and other intra-abdominal infections; septicaemia, skin and soft tissue infections; infections of bones and joints; pelvic inflammatory disease, endometritis, gonorrhoea, and other infections of the genital tract.

Prophylaxis: (Cefoperazone Sodium may be indicated in the prophylaxis of postoperative infection in patients undergoing abdominal and gynecological surgery, cardiovascular and orthopedic surgery).

Combination Therapy: Because of the broad spectrum of activity of Cefoperazone most infections can be treated adequately with this antibiotic alone. However, Cefoperazone may be used concomitantly with other antibiotics if such combinations are indicated. If an aminoglycoside is used, renal function should be monitored during the course of therapy (see Dosage and Administration).

RECOMMENDED DOSAGE:

Dosage and Administration: The usual adult daily dosage of cefoperazone is 2-4 g/day administered in equally divided doses every 12 hrs. In severe infections the dosage may be increased to a total of 8 g/day in equally divided doses every 12 hrs. 12 g/day has been administered in equally divided doses every 8 hrs and usage of up to 16 g/day in divided doses has been reported without complications. Treatment may be started before results of susceptibility testing are available.

The recommended dosage for uncomplicated gonococcal urethritis is 500 mg IM as a single dose. Because renal excretion is not the main route of elimination of cefoperazone patients with renal failure require no adjustment in dosing when usual dosages (2-4 g daily) are administered.

For patients whose glomerular filtration rate is <18 mL/min or whose serum creatinine level is > 3.5 mg/dL, the maximum dosage of cefoperazone should be 4 g/day.

Solutions of Cefoperazone and aminoglycoside should not be directly mixed, since there is a physical incompatibility between them. If combination therapy with cefoperazone and an aminoglycoside is contemplated (see Indications) this can be accomplished by sequential intermittent IV infusion provided that separate secondary IV tubing is used and that the primary IV tubing is adequately irrigated with an approved diluent between doses. It is also suggested that cefoperazone be administered prior to the aminoglycoside.

In infants and children, a 50-200 mg/kg/day dosage of cefoperazone should be given in 2 administration (every 12 hrs) or more if necessary. For neonates aged < 8 days, the drugs should be given every 12 hrs. A dosage of up to 300 mg/kg/day has been used to treat some infants and children with severe infections, without complication.

For the antibiotic prophylaxis of surgical procedures 1 or 2 g should be administered IV 30-90 mins. prior to the start of surgery. The dose may be repeated every 12 hrs for, in most cases, no longer than 24 hrs. In surgery where the incidence of infection is known to be greater (eg. colorectal surgery) or when the occurrence of infection may be particularly devastating (eg. open heart surgery and prosthetic arthroplasty), the prophylactic administration of cefoperazone may be continued for 72 hrs following the completion of surgery.

IV administration: Vials of cefoperazone sterile powder may be initially reconstituted with a minimum of 2.8 mL/g of Cefoperazone of any compatible reconstituting solution appropriate for IV administration listed in Table 2. For ease of reconstitution, the use of 5 mL of compatible solution per g of cefoperazone is recommended (see Table 2).

Table 2. SOLUTIONS FOR INITIAL RECONSTITUTION	
5% Dextrose Injection	10% Dextrose Injection
5% Dextrose and 0.9% Sodium Chloride Injection	0.9% Sodium Chloride Injection
5% Dextrose and 0.2% Sodium Chloride Injection	Normosol M and 5% Dextrose Injection
	Normosol R
	Sterile Water for Injections

The entire quantity of the resulting solution should then be further diluted for administration using any of the following vehicles for IV infusion (see Table 3).

Table 3. VEHICLES FOR IV INFUSION	
5% Dextrose Injection	10% Dextrose Injection
5% Dextrose and Lactated Ringer's Injection	Lactated Ringer's Injection
5% Dextrose and 0.9% Sodium Chloride Injection	0.9% Sodium Chloride Injection
5% Dextrose and 0.2% Sodium Chloride Injection	Normosol M and 5% Dextrose Injection
	Normosol R

For intermittent IV infusion each 1- or 2 g vial of Cefoperazone should be dissolved in 20-100 mL of a compatible sterile IV solution and infused over a period of 15 min.-1 hr. If sterile water for injection is the preferred diluent, no more than 20 mL should be added to the vial.

Table 4. FOR IM/IV				
Vial (g)	Final Cefoperazone concentration (mg / mL)	Step 1 Volume of Sterile Water (mL)	Step 2 Volume of 2% Lidocaine (mL)	Withdrawable Volume (mL)
0.5	250	1.3	0.4	2.0
	333	0.9	0.3	1.5
1.0	250	2.6	0.9	4.0
	333	1.8	0.6	3.0
2.0	250	5.2	1.8	8.0
	333	3.7	1.2	6.0
*There is sufficient excess present to allow for withdrawal and administration of the stated volume				

For continuous IV infusion, each gram of cefoperazone should be dissolved in either 5 mL of Sterile Water for Injection or Bacteriostatic water for injection and the solution added to an appropriate IV diluent. For direct IV injection, the maximum dose of cefoperazone should be 2 g/day administration for adults and 50 mg/kg administration for children. The drug should be dissolved in an appropriate diluent to give a final concentration of 100 mg/mL and administered over a period of no less than 3-5 min.

IM administration: Sterile Water for Injection or Bacteriostatic Water for Injection may be used to prepare Cefoperazone for IM injection. When concentrations of 250 mg/mL or more are to be administered, a lidocaine solution should be used. These solutions should be prepared using a combination of Sterile Water for Injection and 2% Lidocaine HCl Injection that approximates a 0.5% Lidocaine HCl solution. A 2-step dilution process as follows is recommended:

First, add the required amount of Sterile Water for Injection and agitate until Cefoperazone powder is completely dissolved.

Second, add the required amount of 2% Lidocaine and mix (see Table 4 above).

The drug should be given by deep IM injection into the large muscle mass of gluteus maximum or anterior thigh.

ROUTE OF ADMINISTRATION: For i.m. and i.v injection.

CONTRAINDICATIONS:

Known allergy to the cephalosporin class of antibiotics. Not to be used in patients with known hypersensitivity to penicillins.

WARNINGS AND PRECAUTIONS:

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Bicafer Sterile 1g, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Bicafer Sterile 1g must be discontinued immediately and appropriate alternative therapy instituted.

Cefoperazone sodium should not be given to patients who are hypersensitive to it or other cephalosporins. About 10% of penicillin-sensitive patients will also be allergic to cephalosporins and therefore great care should be taken if cefoperazone is given to patients who are known to be allergic to penicillins.

Care is also necessary in patients with known histories of allergy. Cefoperazone sodium should be given with caution to patients with renal impairment; a dosage reduction may be necessary. The concomitant use of diuretics such as frusemide and nephrotoxic antibiotics such as gentamicin may increase the risk of kidney damage. Renal and haematological status should be monitored especially during prolonged and high-dose therapy.

A disulfiram-like interaction with alcohol may occur, and patients receiving cefoperazone sodium should therefore avoid alcohol during and for at least several days after treatment.

Caution: The admixture of penicillins and cephalosporins and aminoglycosides may result in substantial mutual inactivation.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Cefoperazone sodium interacts with alcohol (disulfiram-like reaction), oral anticoagulants and other medications that affect blood clotting. It does not interact with probenecid.

USE IN PREGNANCY AND LACTATION:

Pregnancy and the Neonate: The view that cephamandole and other cephalosporins containing an N-methyltetrazole side chain should be avoided in pregnancy on theoretical grounds of interference with vitamin K metabolism.

SIDE EFFECTS:

Hypoprothrombinaemia has been reported in patients treated with cefoperazone sodium and has rarely been associated with bleeding episodes. Prothrombin time should be monitored in patients at risk of hypoprothrombinaemia and vitamin K administered if necessary.

Allergic reactions, including skin rashes, urticaria, eosinophilia, fever, reactions resembling serum sickness, and anaphylaxis may occur in patients receiving cefoperazone sodium, especially if they are hypersensitive to penicillins or other cephalosporins.

Neutropenia, leucopenia, thrombocytopenia, and haemolytic anaemia have occasionally been reported. Nephrotoxicity has occurred particularly in patients with existing renal impairment or in patients receiving concomitant nephrotoxic drugs. Transient increases in liver enzyme values, including serum aspartate aminotransferase and alkaline phosphatase, have been reported.

Neurological disturbances including encephalopathy have occurred occasionally. There may be pain at the injection site following intramuscular and thrombophlebitis has occurred following intravenous infusion, usually of more than 6 g daily for more than 3 days.

Gastro-intestinal adverse effects have been reported rarely. Prolonged use may result in overgrowth of non-susceptible organisms and as with other broad-spectrum antibiotics, pseudomembranous colitis may develop.

Pseudomembranous colitis – severe abdominal or stomach cramps and pain; abdominal tenderness; watery and severe diarrhoea, which may be bloody and fever.

SYMPTOMS AND TREATMENT OF OVERDOSE:

No known toxicities due to an overdose of cephalosporins in general.

INCOMPATIBILITIES:

Incompatibility or loss of activity has been reported between cefoperazone sodium and aminoglycosides, perphenazine, or pethidine hydrochloride.

STORAGE CONDITIONS:

Store below 25°C. Protect from light.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

Refer under dosage for dilution, etc.

A solution of 300 mg/ml is reported to be stable for 24 hours at 15°C to 25°C and 5 days at 2°C to 8°C.

PACK SIZES:

Pack of 1 g in 1's, 10's 25's, 50's and 100's vials of 10 ml / box.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59, Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA